

Aspiring Quantitative Researcher | Machine Learning Engineer | Corporate Finance

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Professional Summary

Second-year Computer Science & Engineering (Data Science) student building valuation models, trading signals, and market research. Interested in quantitative finance and machine learning.

Technical Skills

- **Programming:** Python, C++, C, SQL, JavaScript
- **Machine Learning:** Scikit-Learn, XGBoost, Feature Engineering, Time Series Forecasting, Model Evaluation, Cross Validation
- **Data Engineering:** Pandas, NumPy, SQL, Power BI, ETL Pipelines, Flask, Streamlit
- **Corporate Finance:** DCF, Financial Statement Analysis, Business Valuation, Investment Analysis, Capital Budgeting, Restructuring Models
- **Quantitative Finance:** Market Regime Analysis, Backtesting, Risk Management, Portfolio Construction, Factor Investing, Monte Carlo Simulation
- **Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Projects

DCF Valuation & Scenario Engine

- NSE valuation terminal: ten-year DCF, Monte Carlo, sensitivity, peer comparison. Python, Flask, SciPy.

Hybrid Stock Prediction Model

- Nifty 50 scanner scoring stocks from ML, DCF, flow, and regime. Team project.

AI-Powered Financial Digital Twin

- Ridge next-day forecast plus a bounded LLM adjustment. Python, Streamlit.

Institutional Flow Tracker

- Accumulation score from public exchange data. Synthetic data only. Python, PCA.

Spirit Airlines Ch. 11 Waterfall Model

- Excel model: 13-week liquidity, DIP sizing, priority waterfall.

Research (Unpublished)

- **Volatility-Uncertainty Dynamics:** Monte Carlo framework and Uncertainty Index. Sole author.
- **Smart Meter Anomaly Detection:** Energy theft detector, synthetic benchmark. Co-authored.

Experience & Certifications

- JPMorgan Investment Banking Virtual Experience
- Goldman Sachs Risk Virtual Experience
- Financial Markets Yale University
- OSI Cloud Infrastructure Oracle
- Machine Learning For Trading Quantinsti
- IBM Data Science Professional Certificate (In Progress)

Education

B.Tech: SY CSE(DS)- Vishwakarma Institute of Technology, Pune

HSC: Shiv Chhatrapati College, Chhatrapati Sambhajinagar

SSC: The Jain International School, Chhatrapati
Sambhajinagar